NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA-136119
B. Tech. (PIE) – 3rd Sem.; Mid-SEM Exam –II
November-2025

PIPC - 202 **Fluid Mechanics & Machines**
Time: 50 Minutes ***Max. Marks: 15***

Note: Attempt all questions. Assume suitable missing data, if any.

Q-1: A jet of water of diameter 6.5 cm strikes a curved plate at its centre with a velocity of 24 m/s. The curved plate is moving with a velocity of 6 m/s in the direction of the jet. The jet is deflected through an angle of 158°. Assuming the plate is smooth, find:
(i) Force exerted on the plate in the direction of the jet,
(ii) Power of the jet, and (iii) Efficiency of the jet. **(4)**

Q-2: Compare and contrast the Pelton, Francis, and Kaplan turbines based on their working principle, head, flow type, shaft position, specific speed, efficiency, and applications. **(3)**

Q-3: Explain the various performance characteristic curves of a centrifugal pump, including the head, discharge, power, and efficiency characteristics. Discuss their practical significance in pump selection and operation. **(3)**

Q-4: *A one-fifth scale model of a pump was tested in a laboratory at 1000 r.p.m. The head developed and the power input at the best efficiency point were found to be 8 m and 30 kW respectively. If the prototype pump has to work against a head of 25 m, determine its working speed, the power required to drive it and the ratio of the flow rates handled by the two pumps.* **(5)**

*****End of Question Paper*****